

ASSIGNMENT-4

Problem Statement: Design and Implementation of a 4-Way Traffic Light Controller Using Verilog HDL

Design and implement a Finite State Machine (FSM)-based 4-Way Traffic Light Controller in Verilog HDL for managing traffic at a road intersection consisting of North-South (NS) and East-West (EW) directions.

The controller shall operate according to the following sequence:

1. North-South Green, East-West Red
2. North-South Yellow, East-West Red
3. North-South Red, East-West Green
4. North-South Red, East-West Yellow
5. Repeat the cycle continuously.

The design must include:

- A clock (clk) input.
- A synchronous or asynchronous reset (rst) input.
- Separate 3-bit outputs for North-South and East-West traffic lights.
- FSM-based state transitions to control the traffic signal sequence.

Inputs:

Signal	Description
clk	System clock
rst	Reset signal

Outputs:

Signal	Description
ns_light[2:0]	North-South traffic light status
ew_light[2:0]	East-West traffic light status

Light Encoding:

Red = 3'b100

Yellow = 3'b010

Green = 3'b001

